

LOW POWER CRYPTO-CHIP FOR IOT APPLICATIONS

Secure IoT demands ultra-low power cryptography solutions. The PRESENT algorithm offers a lightweight, hardware-friendly block cipher perfect for resource-constrained devices.

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INTRODUCTION

- Low power crypto chips are specially designed to perform encryption and decryption efficiently with very low energy usage.
- These chips use lightweight cryptographic algorithms that are optimized for speed and energy efficiency.
- They enable secure communication in IoT devices across various fields like smart homes, healthcare, agriculture, and industrial systems.
- Such chips help achieve real-time security while maintaining long battery life in IoT nodes.



PROJECT OBJECTIVES



Design Implementation

Create hardware-optimized PRESENT cipher implementation for IoT applications.



Power Optimization

Achieve sub-10 μ W power consumption during active encryption operations



Performance Testing

Measure throughput, latency, and energy consumption on target hardware



Security Validation

Verify resistance against side-channel and cryptanalytic attacks.



PROBLEM STATEMENT

IoT devices face unique challenges that conventional cryptographic solutions fail to address.



Power Constraints

Battery-powered devices need years of operation without replacement.



Area Limitations

Minimal silicon area available for security functions.



Security Needs

Must resist sophisticated attacks despite limited resources.

KEY FEATURES OF PRESENT ALGORITHM



Lightweight Design

64-bit block size with 80/128-bit key options.
Requires just 1570 gate equivalents



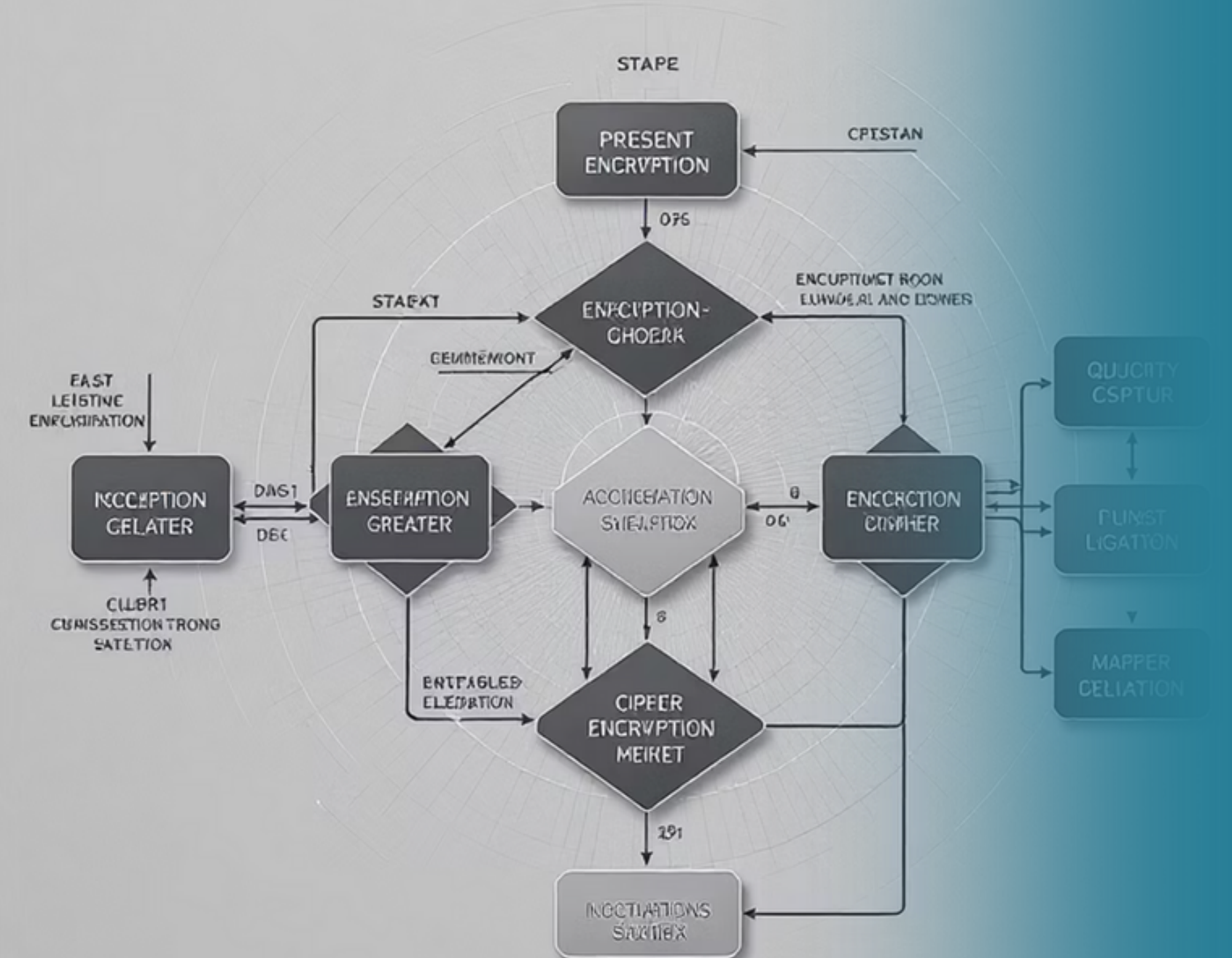
Energy Efficiency

Optimized for hardware implementation with
minimal power consumption per operation



Security Strength

31 rounds provide adequate security margin
against known attacks for IoT applications.



Overview: PRESENT Algorithm

Key Features

Structure

- Lightweight block cipher with ISO/IEC standardization
- 64-bit block size with 80/128-bit key options
- Optimized for hardware-constrained environments
- 31 rounds of substitution-permutation network

Algorithm

Key Addition

XOR operation with round key

S-Box Layer

Non-linear substitution with 4-bit S-boxes

Permutation Layer

Bit permutation for diffusion

Challenges in IoT Security



Resource Limitations

Traditional crypto demands too much power and memory.

Upgrade Difficulties

Long-lived devices need cryptosystem refreshes.

Quantum Threats

Emerging quantum computing poses additional security risks.

Performance & Results: Measured Metrics

Throughput

Up to 200 Mbps @ 100 MHz on
FPGA implementation

Security

Level

Resistant to differential
cryptanalysis attacks up to 16
rounds



Energy Efficiency

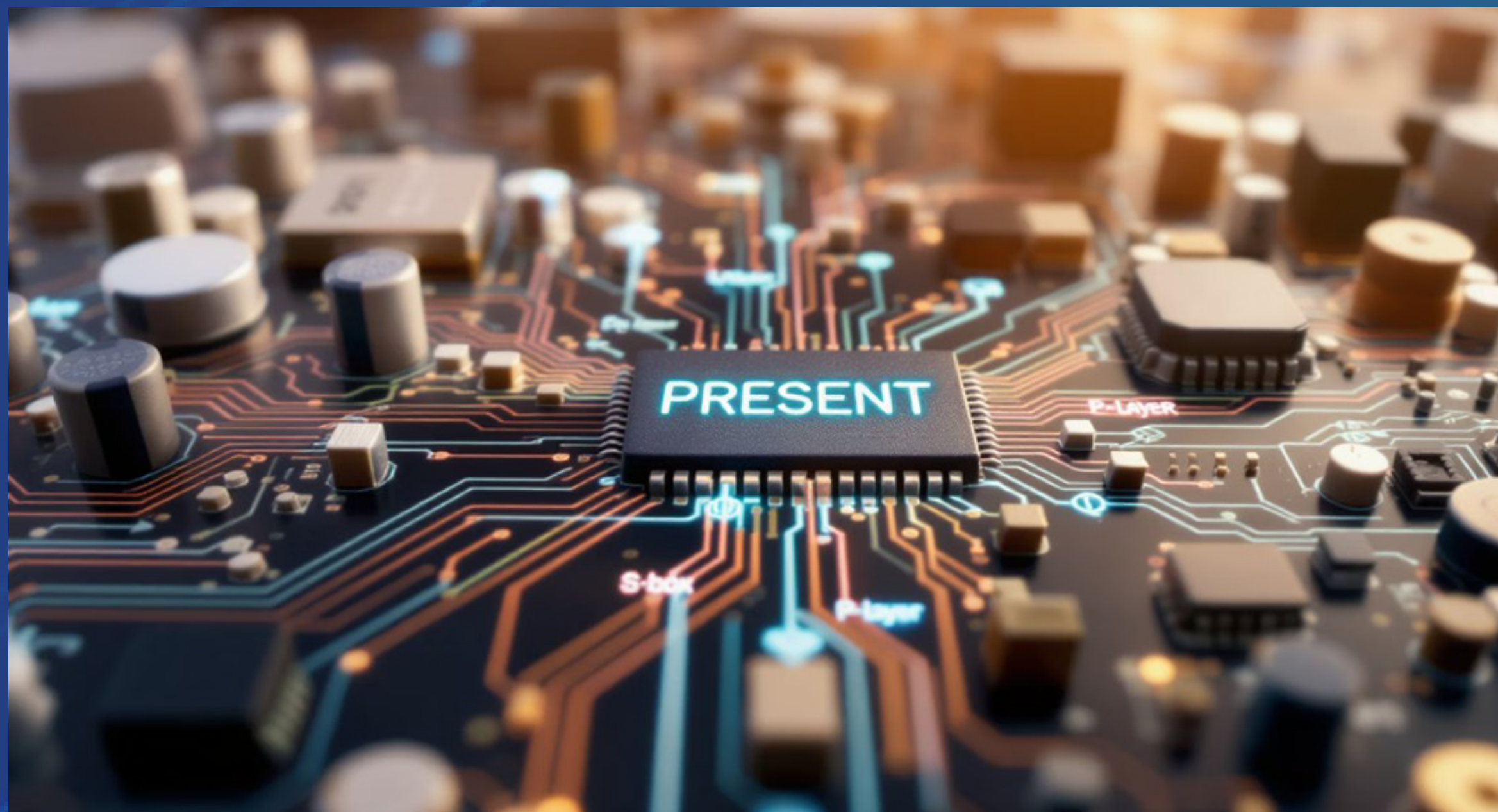
50-200 pJ per bit on measured hardware
chips

Area Efficiency

1.5K-2.5K gate equivalents depending
on optimization

Sample PRESENT Algorithm Implementation (Verilog)

This hardware implementation illustrates the core PRESENT algorithm components



```
module present_round (input [63:0]
    state_in,
    input [79:0] key,
    output [63:0] state_out
);
    wire [63:0] after_keyadd;
    wire [63:0] after_sbox;

    // Key addition
    assign after_keyadd = state_in ^
    key[79:16];

    // S-box layer present_sbox sbox_inst (
        .data_in(after_keyadd),
        .data_out(after_sbox)
    );

    // Permutation layer present_perm
    perm_inst (
        .data_in(after_sbox),
        .data_out(state_out)
    );
endmodule
```

Conclusion & Future

PRESENT algorithm offers a proven solution for low-power, secure IoT hardware implementations.

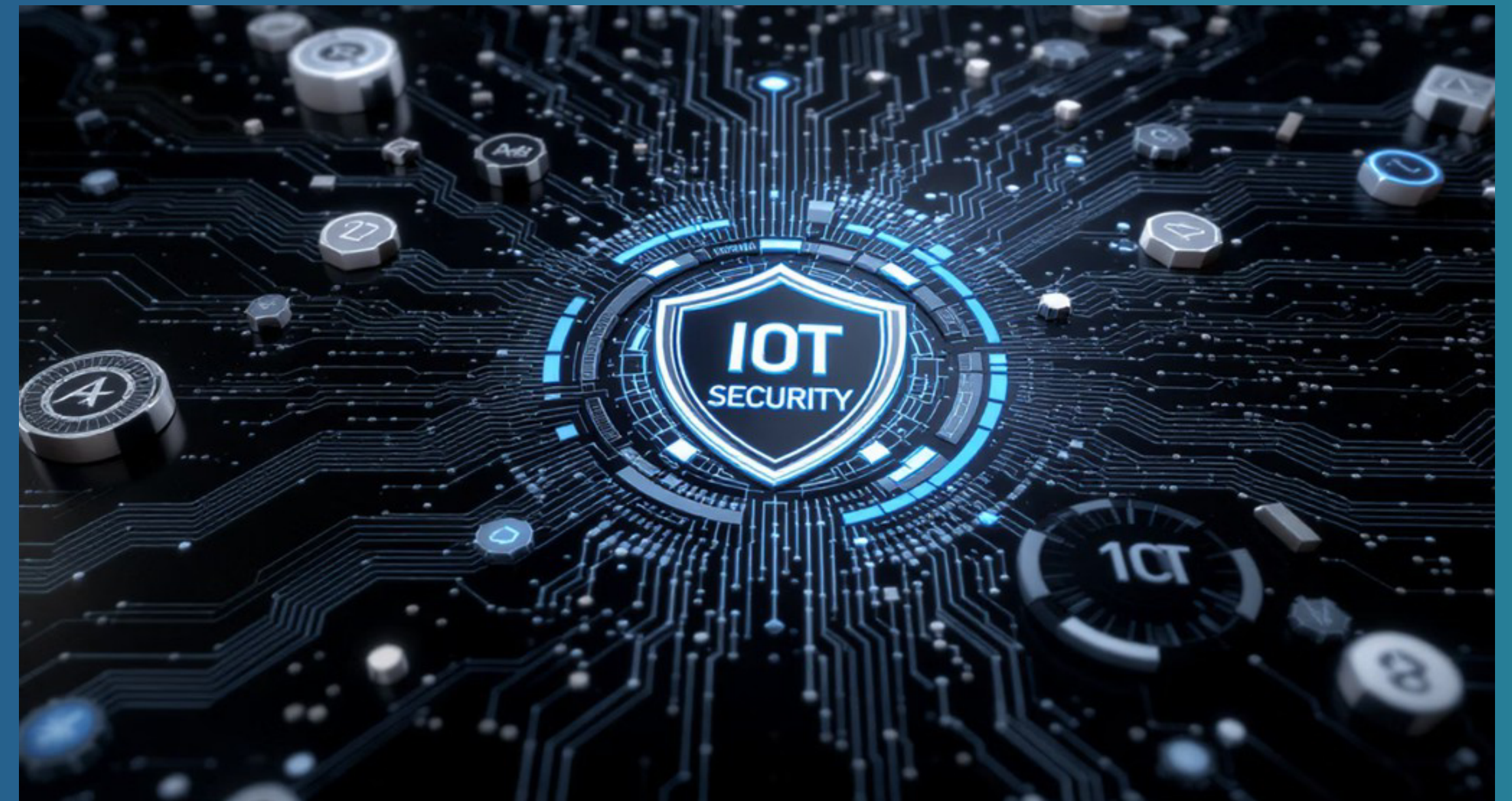
Hardware crypto-chips significantly extend battery life while simplifying software integration.

Future Directions

Post-quantum cryptography adaptations for IoT security

Emerging Technologies

Reconfigurable and AI-accelerated cryptographic workflows





THANK YOU